

Patrick Li-Yu Lo

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Research Dr, Durham NC



<https://pattylo.github.io/>

BACKGROUND

Duke University

PhD Student in Robotics (Mechanical Engineering)

Aug '25 - Present

GPA: 3.88/4.0

The Hong Kong Polytechnic University

MPhil in Robotics & Control (Presidential Fellow)

Sep '22 - Mar '25

GPA: 4.0/4.3

Hong Kong Center for Construction Robotics (HKCRC)

Research Assistant

Jul '21 - Jul '22

The Hong Kong Polytechnic University

BEng (1st Hons) in Aeronautical and Aviation Engineering (APEC Entry Scholarship)

Sep '17 - Jun '21

GPA: 3.75/4.3

The University of Queensland, Australia

Exchange Student

Feb '20 - Jul '20

PUBLICATIONS & PATENTS

First-Author Publications

- **L.-Y. Lo**[†], Y. Liu[†], C. Shu, T. Harris, J. Ryan, and B. Chen, “Fly, drive, reconfigure: A modular reconfigurable aerial-ground platform for field operations,” *arXiv preprint arXiv:2609.28887*, 2026. *Links: pdf, code.*
- N. Weaver[†], B. Xia[†], **L.-Y. Lo**[†], Y. Huang, and B. Chen, “Omnidirectional amphibious locomotion via internal mass actuation,” *arXiv preprint arXiv:2609.27358*, 2026. *Links: pdf, code.*
- **L.-Y. Lo**, B. Li, C.-Y. Wen, and C.-W. Chang, “Experimental non-robocentric dynamic landing of quadrotor UAVs with on-ground sensor suite,” *IEEE Transactions on Instrumentation and Measurement (TIM)*, vol. 73, 2024. *Links: pdf, code.*
- **L.-Y. Lo**, Y. Hu, B. Li, C.-Y. Wen, and Y. Yang, “An adaptive model predictive control for unmanned underwater vehicles subject to external disturbances and measurement noise,” in *2024 14th Asian Control Conference (ASCC)*. IEEE, 2024, pp. 01–07. *Links: pdf, code.*
- **L.-Y. Lo**, B. Li, C.-Y. Wen, and C.-W. Chang, “Landing a quadrotor on a ground vehicle without exteroceptive airborne sensors: A non-robocentric framework and implementation,” in *2023 IEEE 26th International Conference on Intelligent Transportation Systems (ITSC)*. IEEE, 2023, pp. 6080–6087. *Links: pdf.*
- **L.-Y. Lo**, C. H. Yiu, Y. Tang, A.-S. Yang, B. Li, and C.-Y. Wen, “Dynamic object tracking on autonomous UAV system for surveillance applications,” *Sensors*, vol. 21, no. 23, p. 7888, 2021. Editor’s Choice Article. *Links: pdf, code.*

Other Publications & Patents

- Y. Liu, R. Prakash, **L.-Y. Lo**, N. Roede, and B. Chen, “Embodied passive aeroacoustic perception enables relative sensing and pursuit between aerial robots,” *arXiv preprint arXiv:2608.00401*, 2026. *Links: pdf, code.*
- Y. Yang, K. Liu, **L.-Y. Lo**, T. Huang, Y. Fu, and C.-Y. Wen, “Fixed-time adaptive consensus control for multi-quadrotor subject to external disturbances via deep reinforcement learning,” *Aerospace Science and Technology (AST)*, p. 111133, 2025. *Links: pdf.*
- W. Yang, K. Liu, Z. Tan, **L.-Y. Lo**, Y. Wang, K.-H. Wong, and C.-Y. Wen, “Hierarchical 3D scene graph based semantic-metric SLAM for plant inspection and fruit counting in intelligent hydroponics system,” *IEEE Internet of Things Journal*, 2025. *Links: pdf.*
- C.-W. Chang, L.K. Chung, W.-C. Hung, and **L.-Y. Lo**, “An integrated auxiliary system for construction precast components installation,” Patent Pending, Application No.: 202310478194.5, *CN Patent*, [Filing Date: 2023-08-29].

AWARDS & SCHOLARSHIP

- First Runner-up of UAV Challenge, '23, '24, & '25 IEEE International Conference on Unmanned Aircraft Systems (ICUAS). *Links: web '23, web '24, web '25.*
- PolyU Presidential Postgraduate Fellowship Scheme ('22 - '24).

- Dean's List of PolyU Faculty of Engineering: '17/18, '18/19 & '20/21.
- PolyU Undergraduate APEC Entry Scholarship ('17 - '21).
- Wong Tit-shing Student Exchange Scholarship '19/20.
- Dr. Y.K. Ching Memorial Scholarship '19.

CORE SKILLS & EXPERIENCES

Languages & Tools: C++, Python, ROS 1/2, Matlab, Docker, Arduino, PyTorch, & OpenCV.

Expertise: Nonlinear and Convex Optimization, Dynamics & Control, Data-Driven Methods, Opt-Based SLAM.

Robotic Platforms: PX4/ArduPilot, AgileX Scout-Mini, BlueROV2, CrazyFly, UniTree Go1.

Links: [Project Pages](#), [Learning Notes](#).

OTHERS

- **Languages:** English (C1), Mandarin (Native), Cantonese (Proficient), Taiwanese (Proficient).
 - **Reviewing:** CoRL '26, IEEE ITSC '23; IEEE TIM, TMECH, TVT, Sensors Journal; Nonlinear Dynamics.
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LEADERSHIP & SERVICE

- **Leadership**
 President, Duke Taiwanese Student Association '26–'27
 Student Ambassador, HKPolyU '19–'21
 - **Teaching Assistant**
 ME555 Robot Manipulation, Duke University Sep '26
 ENG2002 Computer Programming, HKPolyU Sep '23
 AAE1BN01 Introduction to Aviation Industry, HKPolyU Jun '23
 - **Volunteer Teaching Assistant**
 HeartFire Education Service, China Dec '19
 African Evangelistic Enterprise, Rwanda Jun '19
 Royal University of Phnom Penh, Cambodia May–Jun '18
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REFERENCES

Prof. Chih-Yung Wen	Dr. Boyang Li	Dr. Li-Ta Hsu	Dr. Ching-Wei Chang
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